Advanced Programming Practice · 21CSC203P

Unit 3 · Session 8 · SLO 1 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 3 : Session â€“ 8 : SLO â€“ 1 | Apply Exception Handling in Database Access

I. Fill in the Blanks

1. SQLException â€” it is the base checked exception for all JDBC database access errors.
2. finally â€” the finally block (or try-with-resources) ensures database resources are closed properly.
3. AutoCloseable â€” try-with-resources automatically closes any object that implements the AutoCloseable interface.
4. subclassâ€“superclass (inheritance) â€” in a multi-catch block the exception types must not be related by inheritance.

II. True or False

1. TRUE â€” SQLException provides getMessage(), getSQLState() and getErrorCode() for diagnosing database errors.
2. TRUE â€” ResultSet and Statement must be explicitly closed in a finally block, or automatically via try-with-resources.

III. Multiple Choice

1. B â€” catch(SQLException e) is the correct syntax for catching database-related exceptions.
2. C â€” getSQLState() returns a standardized five-character SQL error code string.

IV. Programming

1. Identify the output: A â€” The connection fails, the SQLException is caught, and the program prints "Error: " followed by the driver-supplied message (e.g. invalid URL / connection refused).
2. Identify the error: B â€” executeQuery() throws SQLException, not IOException; the catch clause must declare SQLException (or Exception), otherwise the program does not compile.